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FACULTY OF AGRICULTURE AND ENVIRONMENTAL SCIENCES

BACHELOR OF SCIENCE IN AGRICULTURE 2025

AGR 1103 Introductory Microbiology Test 01

DATE: 16th October 2025

Student's Name & Reg number KISEKKA PAUL KAGGWA
2025/A/KSA/019/G/F

INSTRUCTIONS:

1. **Attempt all questions; Time allowed ONE HOUR**

Question 01:

a) Define the following terms as applied to microbiology

i) A microbe **(01 mark)**

This is a small organism that is not visible to the naked eye

ii) Pure culture **(01 mark)**

This is a population of microbes arising from a single cell and belong to the same species.

iii) Culture medium **(01 mark)**

This is a composition of nutrients that can allow microbial growth.

iv) A microbial species **(02 marks)**

This is a collection of strains that show similar stable properties

v) Microbial growth **(01 mark)**

This is the increase in the number of microbial cells

b) (i) State the **theory of sponteneous generation** applied to microbiology. **(01 mark)**

States that microbes arise from non-living matter

(ii) Outline the 4 Kotch's postulates (02 marks)

A microbe must be present in the sick animal but absent in the healthy ones

02 The microbe should be isolated ~~and~~ from the sick animal and grown as pure culture

The microbe must cause the disease in a healthy animal once inoculated into it showing the same symptoms

The microbe must be re-isolated and cultured again and it must show similar characteristics

(iii) How are Kotch's postulates relevant in the day today microbiology work? (02 marks)

02 Kotch's postulates provide a systemic framework during which allow microbiologists to get a casual link between the disease and the microbe

archea

Question 02:

a) Explain any five importance of microorganisms on planet earth (05 marks)

05 - Microorganisms have pathogenic properties causing diseases to animals, people and plants.

- Important in industrial work where some are used mainly in brewing industries for fermentation

- Play a role in decomposition of organic matter hence improvement in the soil fertility

Involved in nutrient recycling whereby they play a significant role in the nitrogen and carbon cycles

Useful in production of medicine such as penicilin from the fungi

Some are edible such as mushrooms.

06 b) With relevant examples, differentiate between Prokaryotic microbes and Eukaryotic microbes (06 marks)

Prokaryotic microbes are usually unicellular like the bacteria while eukaryotic microbes are multicellular like the fungi (except yeast)

Prokaryotic microbes like bacteria lack membrane bound organelles while the Eukkaryotic microbes like fungi have membrane-bound organelles

Prokaryotic microbes like archea have a naked DNA within the cytoplasm while eukaryotic microbes like fungi have an enclosed DNA within the nucleus.

15

c) Differentiate the following terms as applied to microbiology

02

i) Anaerobic respiration and Fermentation **(02 marks)**

Anaerobic respiration involves the electron transport chain whereas fermentation does not involve the electron transport chain

02

ii) Catabolism and Anabolism **(02 marks)**

Catabolism is the process that involves breakdown of complex molecules into smaller molecules while

Anabolism is the process that involves the formation of complex molecules from simple molecules.

02

iii) Photoautotrophic microbes and Chemoheterotrophic microbes **(02 marks)**

Photoautotrophic microbes are small organisms that derive their energy from the sun and carbon from carbondioxide directly while

Chemoheterotrophic microbes are small organisms that derive their energy from oxidation of inorganic chemicals and obtain carbon from other organic sources other than carboondioxide.

02

iv) Mycerial strands and Rhizomorph **(02 marks)**

Mycerial strands are aggregations of parallel or interwoven undifferentiated hyphae which are cemented together while

Rhizomorph are aggregations of differentiated hyphae having well defined apical meristems

Question 03:

04

d) Outline any 04 factors that affect growth of microbes e.g. fungi and bacteria **(04 marks)**

Temperature

pH of the medium

Tonicity of the medium.

Radiation

Nutrient requirement

07

e) With respect to the life cycle and structure, differentiate between the following:

i) Zygomycetes and Oomycetes **(02 marks)**

Zygomycetes produce zygospores whereas oomycetes produce oospores

Zygomycetes produce their spores in sporangia while the oomycetes produce their spores in oogonia

Zygomycetes produce non-motile spores while oomycetes produce motile spores.

07

ii) Ascomycetes and Basidiomycetes **(02 marks)**

Ascomycetes reproduce asexually and sexually while the basidiomycetes reproduce primarily using sexual means.

Ascomycetes produce ascospores while basidiomycetes produce basidiospores

14

Question 04

a) Differentiate the following terms as applied to mycology

02 i) Eucarpic fungal thallus and Holocarpic fungal thallus **(02 marks)**
Eucarpic fungal thallus is the fungal body that is differentiated between the vegetative and reproductive parts while
The holocarpic fungal thallus is the one where there is no differentiation between the vegetative and reproductive parts.

02 ii) Mycelium and Hypha **(02 marks)**
Mycelium is a network of fungal hyphae while
Hypha is a tubular, transparent thread-like structure that is either cross walled or not.

02 iii) Lower fungi and Higher fungi **(02 marks)**
Lower fungi have aseptate hyphae while higher fungi have septate hyphae

iv) Perfect fungi and Imperfect fungi **(02 marks)**
Perfect fungi can reproduce both sexually and asexually while
The imperfect fungi reproduce only using asexual means

02 v) Plasmogamy and Karyogamy **(02 marks)**
Plasmogamy involves anastomosis and sharing of the protoplasm between compatible hyphae while
Karyogamy involves the fusion of nuclei from different compatible hyphae to form a dikaryotic hyphae

b) With examples, explain any 04 key features used in classification of lower fungi? **(04 marks)**

02 -Type of spores produced, forexample the zygomycetes produce zygospores and the oomycetes produce the oospores
-The type of flagella posessed either the tinsel-shaped or whiplash forexample the ~~chytrio~~ chytridomycetes have only the whiplash and the oomycetes have both the whiplash and tinsel-shaped flagella
-The mobility of the spores, the oomycetes produce motile spores and the zygomycetes produce non-motile spores.
-Posession of the flagella, the zygomycetes lack the flagella while the oomycetes have the flagella.

Question 05

03 a) List any three different types of asexual spores produced by fungi **(03 marks)**
Conidiospores
Sporangiospores
Chlamydospores

13

b) Explain the three ways by which fungi reproduce **(03 marks)**

Asexual reproduction; this does not involve fusion of gametes or reproductive parts and it involves budding, fragmentation and production of asexual spores

Sexual reproduction: here there's fusion of gametes or reproductive structures and it involves plasmogamy, karyogamy and meiosis.

Parasexual reproduction; the new offsprings are formed ~~in~~ in an almost similar way as in sexual but the stages of plasmogamy, karyogamy and meiosis occur ~~but~~ in a disorganised way

c) Give the implication (s) of each method mention in 4 (b) above to the diversity of fungi **(03 marks)**

Asexual reproduction allows genetic stability and as well reduced hybrid vigour due to reduced genetic diversity

Sexual reproduction decreases genetic stability but increases genetic diversity and hybrid vigour.

Parasexuality also increases genetic diversity

(07)

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DEPARTMENT OF CROP SCIENCE AND PRODUCTION

END OF SEMESTER I EXAMINATIONS 2025/2026

PROGRAMME: BSc. Agriculture

COURSE CODE AND NAME: AGR 1103 Introductory Microbiology

ACADEMIC YEAR: 2025/2026

DATE: 11th December 2025

TIME: 9:00 – 12:00 pm

Instructions;

1. This examination paper consists of SIX questions.
2. All questions carry equal marks
3. Answer any FOUR questions ONLY
4. Do not start answering questions until you are told to do so.
5. All rough work should be done in your answer booklet.
6. Begin each question on a fresh page.
7. Number all the questions attempted in the same manner as they appear on the question paper (including the sections and subsections if any)
8. Writing on the question paper is STRICTLY PROHIBITED.
9. Any form of examination malpractice will lead to disciplinary action.

Question 01

a) Define the following terms as applied to microbiology

i) A microbe **(01 mark)**

ii) Pure culture **(01 mark)**

iii) Culture medium **(01 mark)**

iv) A microbial species **(01 marks)**

v) Microbial growth **(01 mark)**

b) (i) State the **theory of spontaneous generation and germ theory** applied to microbiology. **(02 marks)**

(ii) Outline Kotch's postulates **(04 marks)**

(iii) Explain how Kotch's postulates are relevant in the day today microbiology work ? **(04 marks)**

Question 02:

a) Differentiate the following terms as applied to microbiology

i) Mycenal strands and Rhizomorph **(01 mark)**

ii) Anaerobic respiration and Fermentation **(01 mark)**

iii) Lower fungi and Higher fungi **(01 mark)**

iv) Perfect fungi and Imperfect fungi **(01 mark)**

v) Plasmogamy and Karyogamy **(01 mark)**

b) With examples, explain the key features used in classification of lower fungi **(05 marks)**

c) Explain how microbiology as a scientific discipline is important to mankind **(05 marks)**

Question 03:

a) With reference to key characteristics, write short notes on the following plant pathogenic bacteria genera

i) Agrobacterium **(03 marks)**

ii) Eirwinia **(03 marks)**

iii) Pseudomonas **(03 marks)**

iv) Xanthomonas **(03 marks)**

b) Give any one specific disease and its causative agent under each of the following plant pathogenic genera.

i. Ralstonia **(01 mark)**

ii. Pseudomonas **(01 mark)**

iii. Xanthomonas **(01 mark)**

Question 04:

a) Describe any 03 key morphological features used in classification of bacteria during plant diseases diagnosis **(03 marks)**

b) Write short notes on the following terms as applied to bacteriology:

i) Catalase test **(02 marks)**

ii) Urease test **(02 marks)**

iii) Oxidase test **(02 marks)**

c) Explain the morphological basis for classification of bacteria as:

i) Gram positive **and** Gram negative **(03 marks)**

ii) Acid-fast **and** Non-Acid fast **(03 marks)**

Question 05

a) List any 2 types of asexual fungal spores **(01 mark)**

b) With respect to the life cycle and structure, differentiate between the following:

i) Zygomycetes and Oomycetes **(03 marks)**

ii) Ascomycetes and Basidiomycetes **(03 marks)**

iii) Ascomycetes and Deuteromycetes **(02 marks)**

c) Explain the three ways by which fungi reproduce **(03 marks)**

d) Give the implication (s) of each method in 3c above to the genetic diversity of fungi **(03 marks)**

Question 06

a) Differentiate the following terms as applied to virology:

i. Viroid and Virion **(02 marks)**
ii. RNA viruses and DNA viruses **(02 marks)**
iii. Positive sense RNA viruses and Negative sense RNA viruses **(02 marks)**
iv. Single stranded viruses and double stranded viruses **(02 marks)**

b) Explain the importance of viruses in nature and or to man. **(04 marks)**

c) " Viruses are regarded as both non-living and living microbes" Discuss. **(03 marks)**

END